

OzempicAI product strategy and implementation

Executive proposal September 13 2026

Recommendation

Fund a 12 week product program that makes daily nutrition easier for adults managing changing appetite, including people using prescribed GLP-1 medication. First repair incomplete planning and goal controls. Then test an appetite check-in that helps users choose a realistic next meal, remembers their preferences, and turns those choices into a flexible weekly plan. Keep basic tracking useful for people who do not use medication.

Why this direction

The current repository provides authenticated manual logs and daily totals, but little help deciding what to do next. Lifesum already markets photo, voice, text and barcode logging. MyFitnessPal offers medication and side effect tracking, while Noom offers a GLP-1 companion. We should close essential usability gaps and compete on whether our suggestions fit a person's actual day. This positioning is a hypothesis to validate, not a claim of unique market ownership. [S1-S4]

Decision requested

Approve a two week discovery and foundations phase within a provisional 12 week envelope. Assume two full-time engineers, a half-time product designer, a half-time product manager and fractional dietitian and QA support. Release later funding at weeks 2 and 6 against the evidence gates in this document. These are planning assumptions, not a statement about current staffing.

Evidence and limits

The founder reports losing customers to competitors. No production analytics, billing data, support archive or customer interviews were supplied. Findings below come from the local source at commit 4c029f7 and official public competitor documentation accessed September 13, 2026. The deployed app and database were not inspected. Customer segments, estimated effort, pricing and outcome targets are proposals rather than observed results.

What success means

Improve the proportion of eligible users who return in week 4 to complete a useful nutrition action. Use a provisional target of five percentage points of absolute improvement versus a randomized holdout, with no material decline in trust, reliability or paid retention. Establish the baseline and feasible sample size before committing to that threshold.

Current product and competitive context

What exists in this checkout

React 18, TypeScript and Vite provide an installable PWA. Supabase handles authentication and persisted logs, with React Query hooks for data access. Today shows calorie, water and exercise totals. Calories requires a food name and calorie number and supports deletion. Exercise captures duration and optional strength sets and repetitions. Weight and heart rate have dedicated logging screens. [R1–R4]

Where the experience breaks

Plans renders only explanatory placeholder text and refers users to a Mac app that README says was retired. Profile exposes identity and sign-out but explicitly defers daily goal editing. Today falls back to 2,000 kcal and 2,500 ml and fixes exercise at 30 minutes. Food records have no protein, fiber, serving or nutrient provenance fields. These are observable product limitations, not proven churn causes. [R2–R5]

Lifsum establishes the convenience baseline

Lifsum advertises multiple logging modes, meal plans, recipes and food feedback. It is therefore reasonable to expect customers to compare logging effort and planning utility. We should prioritize repeat meals and correction controls before assuming a camera interface alone creates value. Public feature pages do not establish real-world accuracy or competitor retention. [S1, S2]

GLP-1 support is already competitive

MyFitnessPal's official announcement describes medication logs, reminders and side effect monitoring alongside nutrition, available to US users. Noom describes protein and muscle support in its GLP-1 companion and has announced a free version. A medication diary or protein ring can support our experience, but cannot carry our differentiation by itself. [S3, S4]

Strategic choice

Focus initial recruitment on adults who say changing appetite makes meal decisions difficult. Test a daily loop of check-in, suggested meal, confirmation and weekly reflection. Defer a broad recipe marketplace, public social feed, fasting optimization and native wearable integrations until this loop demonstrates value. Review consumer naming and positioning before a broader launch rather than implying affiliation with a medication manufacturer.

Customer discovery and product principles

The priority job

When I have little appetite or my usual meal no longer appeals, help me find something practical I can eat today and remember what worked. This is a proposed job, inferred from the project context and competitive research. It must earn priority through customer evidence.

Three recruitment groups

Interview six recently lapsed users who switched tools, six retained users and six new or struggling users. Within those groups, include medication users and people not using medication, varied food preferences and different levels of tracking experience. Ask participants to reconstruct their last difficult meal decision and demonstrate their current workaround. Do not prompt them to endorse an AI coach.

The first two weeks

With permission, observe eight task sessions covering meal entry, correction, setting goals and planning tomorrow. Use synthetic foods and accounts where practical. Ask switchers what happened immediately before leaving, what they gained elsewhere and what they would lose by returning. Reconcile interview themes with activation funnels and account age. Eighteen interviews reveal patterns, not market prevalence.

Proceed and stop criteria

Proceed with the appetite proposition if at least six of ten recruited target-segment concept testers independently describe the job as frequent and at least seven can choose and save a suitable meal without help. Treat these as qualitative gates, not statistical proof. If faster logging dominates the evidence, move meal capture ahead of personalization. If trust or relevance fails, keep human-curated suggestions and revise the concept.

Experience principles

Every suggestion must be editable or dismissible. Ask for the smallest amount of information that changes the recommendation. Show incomplete nutrition as unknown rather than zero. Let users hide calorie and weight emphasis. Celebrate useful actions without streak loss, public weight rankings or rewards for eating less. Preserve access to personal history and export when a subscription ends.

Evidence collection

Instrument first log, failed save, correction, plan creation, plan use and return visits. Add an optional cancellation reason and a short in-product usefulness question. Never put meal text, symptoms, medication details, email or photos into general analytics. Analyze aggregate cohorts, and seek permission separately for customer research recordings.

Priority features for daily usefulness

F1 My Day and quick repeat

Improve Today into a user-configurable home screen with editable goals, a prominent repeat-last-meal action and entry correction. Example: a user saves a familiar breakfast once, then adds it tomorrow with its previous portion and can change the amount. Replace the Plans placeholder immediately with a useful editable starter plan. This repairs the most obvious reasons to abandon a session.

F1 Technical implementation and release test

Extend ProfileScreen and useUserProfile with a validated update mutation and explicit goal origin. Add useUpdateCalorieLog and a date selector to CalorieScreen. Add nullable nutrient fields to calorie_logs, a favorite_meals table and a client_request_id uniqueness constraint per user. Keep old calorie-only rows valid and display missing macros as unknown. Update Today and useDailyTotals to show coverage. Test saving, editing, deleting and replaying the same request, plus local midnight and daylight-saving boundaries. Target repeat logging under 10 seconds in usability tests. Estimate 2–3 engineer-weeks.

F2 Appetite Compass

Offer an optional check-in for appetite, available time and preferred meal size. Show two or three reviewed meal options that meet stated preferences. A low-appetite user might pick a smaller portion, save the rest for later, or choose another texture. Ask whether the meal worked and make that answer useful next time. Avoid inferring medication effects or lowering nutrition goals from a check-in.

F2 Technical implementation and release test

Add check_ins, recipe_versions, recipe_tags, user_food_preferences and recommendation_feedback. A server function applies allergy exclusions and content eligibility first, then ranks reviewed recipes by explicit preferences, prep time and prior feedback. Store recommendation ID, rules version and reason codes. Start with deterministic rules and 30–50 dietitian-reviewed options. Optional symptom journaling stays separate and never drives treatment advice. Test allergy exclusions, missing goals, sparse history and rejected suggestions. Measure accepted suggestions and perceived usefulness, with W4 retention as the ultimate test. Estimate 3–4 engineer-weeks plus clinical content review.

Why these belong together

Quick repeat makes the app easier to use today. Appetite Compass makes today's experience worth returning to. A technically impressive recommendation engine will fail if logging and correction still feel cumbersome.

Features that make planning flexible

F3 Flexible Week and Rescue Meal

Turn Plans into a three-day editable meal plan with leftovers and an aggregated grocery checklist. A Rescue Meal action replaces just one meal when appetite, time or ingredients change. Preserve other meals and user edits. Example: a dinner requiring 40 minutes becomes a saved 10-minute option, and the grocery delta appears for confirmation. Familiar household food and smaller optional portions are more useful than a rigid perfect week.

F3 Technical implementation and release test

Reuse meal_plans and grocery_items, adding recipe_version_id, servings, plan_version and a meal_plan_items relation. Store structured ingredients with canonical IDs and compatible units. Use a transactional server operation to apply one confirmed swap and recalculate the affected grocery quantities. Do not combine incompatible units or infer household allergies from one user. Offer undo through a plan revision. Preserve purchased items and manual additions. Test concurrent edits, duplicate ingredients, deleted recipes and repeated requests. Measure planned meals actually used, not just plans generated. Estimate 3–4 engineer-weeks, dependent on F1 and the reviewed recipe catalog.

F4 Meal Memory with assisted capture

Let users describe or photograph a meal, review the suggested foods and portions, and save the corrected version for reuse. Personal corrections take precedence the next time. Begin with repeat meals and text-assisted search. Introduce photo input only if it reduces total time including corrections. Voice is a later input option, subject to browser support and actual demand.

F4 Technical implementation and release test

Add food_catalog_refs, meal_items and inference_jobs. A provider-neutral server adapter resolves food candidates against a licensed nutrition source and returns structured estimates with source, portion and uncertainty. The model never writes committed logs. A separate confirmation request saves the reviewed items and a meal total atomically. Keep provider secrets on the server, use short-lived private image access and delete transient media after processing or a documented expiry. Display approximate values and request clarification when portions are ambiguous. Manual capture remains available after timeout. Estimate 4–6 engineer-weeks after F1.

F4 Quality gate

Benchmark on 100 consented or licensed representative meals with reviewed references, including mixed dishes and regional foods. Measure correction time, severe portion errors and nutrient error distributions. Define acceptable error with the dietitian before testing. Gate broader rollout on improved task completion time and no unconfirmed saves. Do not market nutritional precision from a photo.

Features that support lasting progress

F5 Strength and daily life wins

Expand progress beyond the scale with private milestones such as a completed strength session, improved confidence preparing meals or a self-reported daily activity becoming easier. Let users choose what matters and hide weight charts. Offer short reviewed movement routines with ability and equipment filters. This is a hypothesis about motivation, not a claim that the app measures muscle preservation.

F5 Technical implementation and release test

Reuse `exercise_logs`, including its existing sets and reps fields. Add `workout_templates`, `workout_sessions` and `user_milestones`, with optional exertion and exercise load units. Show trends only across comparable exercises. Build a progress view with dates, missing-data states and a user-selected milestone. Test unit conversion and incomplete sessions. Measure repeat participation and usefulness, and avoid equating exercise minutes or weight trends with muscle mass. Estimate 2–3 engineer-weeks after the core release.

F6 Weekly Story and visit brief

Give users a concise weekly reflection explaining what they logged, what felt workable and one optional experiment for next week. Offer a user-reviewed export for a healthcare visit, with nutrition coverage, weight history and optional symptom notes. Example: a user sees that two saved lunches were repeatedly marked helpful, then chooses to reuse one. No diagnosis or causal attribution appears.

F6 Technical implementation and release test

Create a weekly aggregate job over authorized user records and save snapshots in `weekly_reviews` with source row IDs and rules version. Use templates first, adding language generation only for summaries of verified facts. Distinguish unlogged days from zero intake and suppress thin-data conclusions. Generate exports on demand in private storage with short expiry. Let users remove fields before download. Never automatically send reports to clinicians. Test source-to-summary consistency, deleted-record handling and cross-account export denial. Estimate 2–3 engineer-weeks.

F7 Gentle return and maintenance mode

When someone returns after a gap, offer one familiar action and keep their history. A maintenance preference changes emphasis toward routines and confidence instead of continual weight loss. Add user-controlled reminder frequency, quiet hours and pause. Implement `lifecycle_preferences`, `user_action_days` and server-side delivery deduplication. Notifications use neutral text. No medication stopping or tapering recommendations. Measure seven-day return after re-entry against a holdout, plus notification opt-outs. Estimate 1–2 engineer-weeks, with F6 reuse.

Prioritization and delivery capacity

How we chose the order

Rank work by observed product gaps, expected repetition of the customer job, learning value and dependency cost. Confidence is highest for placeholder repairs and lowest for personalization and revenue effects. The estimates below are engineering effort ranges before discovery, including ordinary review and feature tests. Shared foundation work appears once. They are not delivery promises.

Portfolio

Foundation and instrumentation: 2 engineer-weeks, required first. F1 My Day and quick repeat: 2–3, highest confidence. F2 Appetite Compass: 3–4, primary differentiation experiment. F3 Flexible Week: 3–4, closes an existing visible gap. F4 assisted capture: 4–6, competitive convenience with vendor and accuracy risk. F5 strength and wins: 2–3, later motivation experiment. F6 weekly story: 2–3, later retention experiment. F7 gentle return: 1–2, later lifecycle experiment. Full portfolio: 19–27 engineer-weeks before contingency, content work and substantial vendor integration.

Twelve week commitment

Two engineers provide 24 nominal engineer-weeks. Reserve six for hardening, integration, research follow-up and unexpected work, leaving 18 for planned scope. Foundation plus F1–F3 consumes 10–13. Allocate the remaining 5–8 to a limited F4 text-capture beta or additional reliability work. Photo rollout and F5–F7 are conditional follow-on work. A narrow weekly summary prototype can support discovery without committing to production F6.

Dependencies and ownership

Product owns target cohorts, problem validation and release decisions. Design owns observed task completion and accessible flows. Engineer A owns core logging, plans and offline mutations. Engineer B owns data changes, analytics and the recommendation service. The dietitian owns recipe eligibility and nutrition copy. A named release owner coordinates QA, support readiness and kill switches. If clinical review is unavailable, launch logging improvements and reviewed generic planning only.

What we defer

Defer social feeds, challenges based on weight loss, autonomous coaching, medication dose recommendations and extensive food database construction. Native HealthKit access requires a native integration path and cannot be assumed from the existing PWA or the heart-rate source enum. Revisit native work only if discovery shows that integration is a major switching reason. [R1, R4, S8]

Architecture and data implementation

Build on the current stack

Keep React, Vite, React Query and Supabase. Add feature modules for check-ins, recipes, progress and weekly reviews following the existing hooks pattern. Basic authorized CRUD can use the existing Supabase client. Server functions handle external providers, recommendation rules and privileged jobs. A provider failure must never prevent manual logging. Current PWA configuration caches the app shell and fonts but does not implement durable queued health-data writes. [R1, R6, S6]

Data contract

Extend `calorie_logs` additively with `protein_g`, `fiber_g`, `carbs_g`, `fat_g`, `source`, `confirmed_at`, `client_request_id` and an optional `meal_items` relation. Nutrients are nullable nonnegative numerics, with null meaning unknown. Keep calories as the compatible meal total. Store units and serving factors on meal items, plus the source version used at confirmation. Add preferences for timezone, chosen goals, goal source, calorie visibility and reminder consent. Do not silently convert existing default goals into personalized recommendations.

Request contracts

Proposed `POST /meal-estimates` accepts text or a private media reference plus `request_id` and returns candidate foods, quantities, provenance and `requires_confirmation`. `POST /meals/confirm` accepts reviewed items, `logged_at` and `request_id` and returns the committed meal ID. `POST /recommendations` accepts `check_in_id` and returns eligible recipe IDs with reason codes and `rules_version`. `POST /plans/swap` accepts `plan_id`, `expected_version` and a replacement and returns the new version and grocery delta. Derive user identity from verified authentication, never trust a supplied `user_id`.

Consistency and migration

Use a database transaction or narrowly scoped RPC for meal plus item creation and plan swaps. A unique `user_id` and `request_id` pair prevents retry duplicates. Reject stale plan versions with a conflict response. Index user-owned histories by `user_id` and event timestamp. Add schema and policies before deploying readers, backfill only known values, then enable writers behind flags. Keep old clients working through additive fields. Validate totals against original logs and never reconstruct unknown macros from calories.

Offline and time handling

Add an IndexedDB outbox for user-confirmed mutations, scoped to the signed-in user, with explicit pending and failed states. Retry on reconnect and app foreground without relying on background execution. Clear queued and cached sensitive records at logout or account deletion, and prevent another account from replaying them. Preserve the original event timestamp and timezone. Use local calendar keys consistently and test travel, midnight and daylight-saving transitions. Surface stale cached totals rather than presenting them as current.

Trust and release quality

Health boundaries

The product supports tracking and meal decisions. It does not diagnose symptoms, recommend dosing or tell users to stop medication. NIDDK describes medication as part of a broader treatment plan and advises consultation with healthcare professionals. Have a clinician approve symptom wording and escalation paths before any symptom-related feature launches. Persistent or severe concerns must route to appropriate care information, never reassurance from a model. Personalized targets require explicit user or clinician input and reviewed eligibility logic. [S5]

Access and consent

Enable row-level security on every exposed user table with matching grants and ownership checks for each operation. UPDATE requires both row selection and ownership validation of the resulting record. Verify recipe and meal relationships cannot reference another user's private data. Keep service credentials server-side. Test anonymous access, account A accessing account B, storage reads and export downloads. Supabase policies and storage controls require explicit validation, not reliance on frontend filtering. [S7]

Data minimization

Use separate opt-ins for research, optional medication context and external AI processing. Avoid persistent raw photos by default, strip location metadata before upload and document the processing retention window. Proposed transient media deletion is within 24 hours, with immediate deletion after successful processing when possible. Require vendor data-use and retention review before enablement. Account deletion must cover database records, stored files, queued jobs and local caches, with the backup retention policy clearly stated.

Operational quality gates

Before beta, pass critical flows for signup, meal save and correction, plan swap and offline recovery. Test representative iPhone Safari PWA and Android Chrome behavior, keyboard navigation, screen-reader labels, large text and reduced motion. Confirm partial nutrient totals show coverage. Keep sensitive payloads out of logs. Record latency, failure rate, duplicate saves and provider cost using redacted telemetry.

Proposed launch thresholds

Require no unresolved critical privacy or clinical defects, zero duplicates in retry tests and at least 99% successful valid save attempts during beta. Target p95 manual save under two seconds on the agreed test network, excluding offline queuing. Assisted capture has an eight-second timeout with manual fallback. Disable the affected feature immediately for cross-account exposure or harmful advice. Pause expansion for save success below 99% over a meaningful sample or a material correction burden. These thresholds must be finalized against the baseline.

Rollback

Use server-owned feature flags for appetite recommendations, plan swaps and capture providers. Preserve existing logs when disabling a feature. Stop queued provider jobs safely, retain manual entry and make rollback additive-schema compatible. Reconcile pending writes before deleting local state. A rollback must preserve user-created plan edits and downloaded data.

Rollout and measurement plan

Weeks 1 and 2

Recruit discovery cohorts, instrument the existing funnel and repair misleading placeholder copy. Establish save reliability and a retention baseline. Deliver goal editing and repeat-meal prototypes. At the week 2 review, select either the appetite hypothesis or a logging-first focus based on observed user jobs. Produce a reviewed recipe inventory and a data contract before development expands.

Weeks 3 through 6

Ship F1 and basic F3 planning to an internal cohort, then a consented beta of approximately 30–50 target users if recruitment allows. Add the reviewed F2 rules and gather explicit usefulness feedback. Run the appetite flow with manual entry intact. At week 6, require usability and reliability gates before extending assisted capture. Small beta counts are for defect discovery and interviews, not proof of retention lift.

Weeks 7 through 12

Randomize eligible accounts into a stable holdout and treatment group, stratified by new versus existing users. Ramp treatment exposure from 5% to 25% to 50% after each reliability review. Keep a 10% holdout when wider availability begins. Do not ramp merely because a date arrives. Start the retention experiment by week 7 so some cohorts mature before the week 12 decision. If enrollment is slow, extend measurement and report uncertainty.

Primary outcome

W4 useful retention is the fraction of randomized eligible accounts completing a confirmed meal log or marking a planned meal used on at least two distinct local days during days 22–28 after assignment. Count everyone assigned in the denominator, even if they never open the new feature. Assign new users at their first authenticated eligible session. Analyze existing users separately. This is a product-use metric, not evidence of a health benefit.

Supporting and business metrics

Track first-day activation, median time to a confirmed meal, correction rate, used planned meals, suggestion usefulness and support complaints. For paid customers, measure renewal and cancellation separately using verified billing events. No billing integration is visible in the inspected code. Avoid declaring business retention improvement from W4 activity alone. The north-star candidate is weekly users completing a useful nutrition action on at least three days.

Experiment discipline

Use one primary hypothesis, a pre-registered analysis window and account-level assignment. A provisional five-point lift from 25% to 30% requires about 1,250 accounts per arm under a two-sided 5% test with 80% power, before attrition adjustments. This is an illustrative calculation, not our baseline. Recalculate with actual rates and traffic. Log exposure separately from assignment and check sample balance. Use interviews for explanation and avoid repeated significance peeking.

Commercial strategy and investment decisions

Packaging hypothesis

Keep manual logging, editable goals, core history, basic planning and personal data export in the accessible core. Test paid value around convenient capture, richer flexible planning and longer-term insights after users experience a useful result. Do not charge for symptom escalation content or lock people out of their records. Competitor free offerings make an undifferentiated GLP-1 subscription difficult to defend. [S3, S4]

Pricing discovery

No current price, revenue or cancellation cohort was provided. Interview recent switchers about the subscriptions they actually paid for, then test a transparent offer with clear renewal terms. An illustrative \$8–12 monthly range is only a research stimulus, not a recommendation to change prices. Use completed purchase and renewal behavior to validate willingness to pay. Avoid discount campaigns before the underlying experience improves.

Cost envelope

Use 24 engineer-weeks as the 12 week capacity envelope, inclusive of the six-week effort reserve. Illustratively, at a loaded \$4,000 per engineer-week, engineering costs \$96,000. Add \$36,000 for combined product and design effort at one full-time equivalent for 12 weeks at \$3,000 per week, and \$18,000 for clinical content, QA and vendors. The resulting \$150,000 envelope is a planning scenario, not a quote or approved budget. Replace every rate with finance inputs.

Unit economics guardrail

Track AI and food-provider spend per active user and per successfully confirmed assisted meal. As a scenario, 60 assisted meals at \$0.02 each cost \$1.20 per user-month before hosting, support, payment costs and taxes. With \$10 of monthly recognized revenue, that leaves \$8.80 before those other costs. Prices are illustrative, not vendor quotes. Use quotas, caching and a provider kill switch. Evaluate contribution margin using observed usage distribution, especially heavy users.

Decision gates

Week 2: proceed if the target job and first flows validate, otherwise reprioritize logging. Week 6: expand if reliability and relevance pass, otherwise improve the manual and curated experience. Week 12: scale if mature retention evidence supports a worthwhile effect and the experience remains trustworthy. If the confidence interval includes no improvement, keep exposure limited or extend the experiment. Stop the differentiated bet if adequate data excludes a useful effect, even when engagement with the novelty looks strong.

Next decisions for leadership

Confirm the initial audience, available team and real spending envelope. Appoint an owner for customer research and a clinical content reviewer. Approve only the first two weeks now, with subsequent scope governed by the gates above. This document proposes features and implementation work. It does not change production code or authorize a launch.

Sources and evidence register

Repository evidence

Local checkout reviewed September 13, 2026 at commit 4c029f7. Paths are relative to /Users/mihirpatel/Downloads/OzempicAI. R1: README.md and package.json, platform and dependencies. R2: src/features/today/TodayScreen.tsx, current totals and goals. R3: src/features/calories/CalorieScreen.tsx and src/hooks/useCalorieLogs.ts, manual capture and mutations. R4: src/types/db.ts and src/features/exercise/ExerciseScreen.tsx, data model and strength fields. R5: src/features/plans/PlansScreen.tsx and src/features/profile/ProfileScreen.tsx, placeholders. R6: vite.config.ts and src/lib/date.ts, caching and local time handling. R7: supabase/migrations/00005_rls_policies_and_signup_trigger.sql, access policies. Planning notes in web-migration-plan.md informed context but were not treated as proof of implemented features.

S1 Lifesum current product

<https://lifesum.com/>

Official overview of photo, voice, text, barcode and quick tracking. Accessed September 13, 2026.

S2 Lifesum features

<https://lifesum.com/features/>

Official overview of plans, recipes and food feedback. Accessed September 13, 2026.

S3 MyFitnessPal GLP 1 support

<https://news.myfitnesspal.com/myfitnesspal-launches-glp-1-support-to-help-users-stay-consistent-and-build-habits-alongside-medication/>

Official announcement of logging, reminders and side effect monitoring in the US. Accessed September 13, 2026.

S4 Noom companion

<https://www.noom.com/health/glp1companion/>

<https://www.noom.com/in-the-news/noom-launches-free-version-of-its-powerful-glp-1-companion/>

Official product and free-version announcement. Accessed September 13, 2026.

S5 NIDDK treatment context

<https://www.niddk.nih.gov/health-information/weight-management/prescription-medications-treat-overweight-obesity>

Health context for treatment boundaries, not a validated recommendation algorithm. Accessed September 13, 2026.

S6 to S8 Technical references

S6: <https://supabase.com/docs/guides/functions>

S7: <https://supabase.com/docs/guides/database/postgres/row-level-security>

S8: <https://developer.apple.com/documentation/healthkit>

Official architecture and access-control references. Accessed September 13, 2026. Proposed APIs and schemas in this document remain designs and require implementation validation.